Homework Assignment 3A

Power Stage Design

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TU Delft

Part 1

PowerFET Sizing (R_ON <= 50 mOhm at 150C)

Part 1: PowerFET Sizing

Goal: Size the nld12_g5a_nbl_mac PowerFET for R_ON <= 50 mOhm at 150C

Method:

DC sweep of total gate width W

VGS = 5V, VDS = 100mV, T = 150C

R_ON = VDS / ID = 100mV / ID

Result:

At W = 103mm (nf = 400 fingers): R_ON = 50.02 mOhm

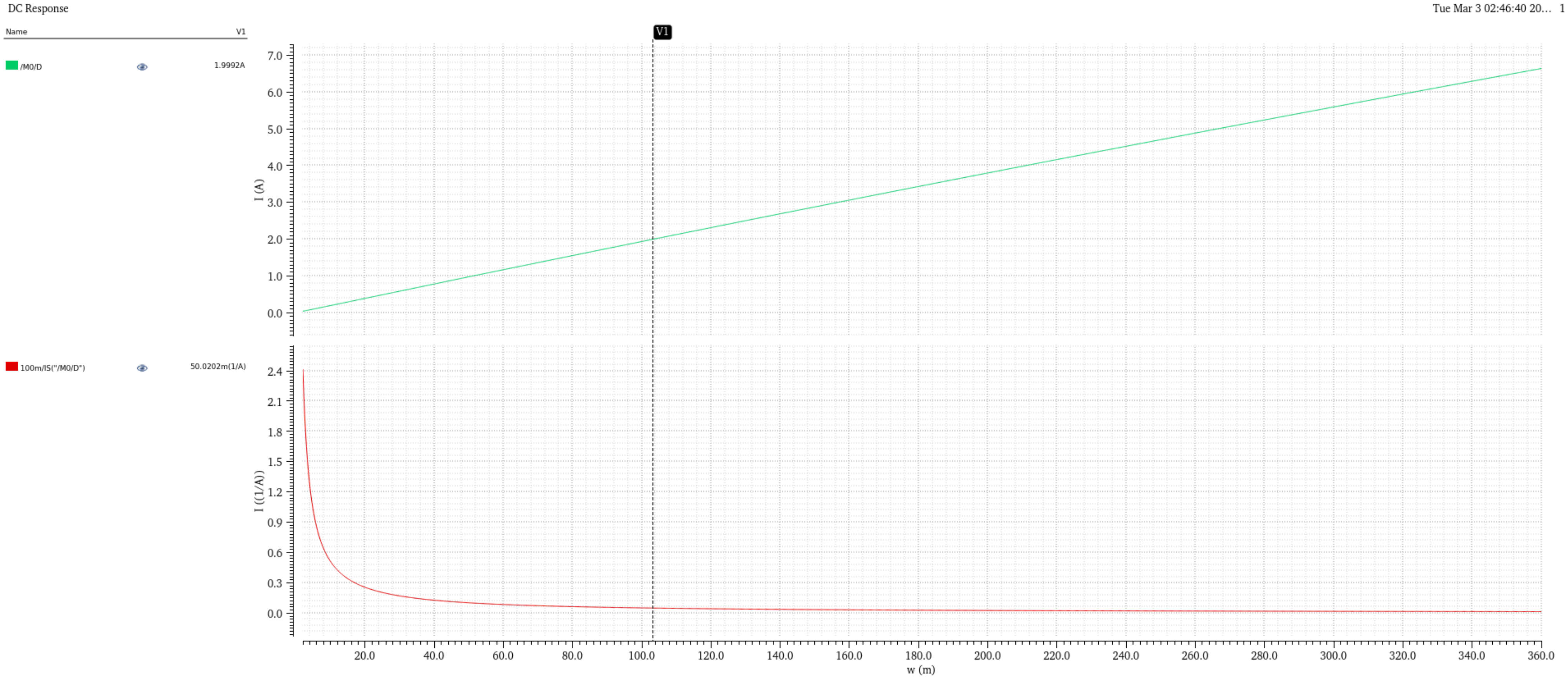
ID = 1.9992A at the target operating point

Key formula:

N_fingers = ceil(R_ON,single / R_ON,target)

W_total = N x W_finger

Part 1: R_ON vs Width



Top: ID vs W, Bottom: R_ON vs W. Marker at W=103mm, R_ON=50.02 mOhm

Part 2

Threshold Voltage & Gate Charge

Part 2: V_TH Extraction & Gate Charge

V_TH Extraction:

DC sweep: VGS from 0 to 5V, VDS = 100mV, T = 27C

V_TH ~ 0.75V (where drain current starts flowing)

Gate Charge Q_G:

Transient simulation: gate driven 0 to 5V

Gate current integrated over time: Q_G = integral(I_G dt)

Q_G = 1.247 nC

Gate charge power loss (per half-bridge):

P_Q = f_SW x 2 x V_SUP x (Q_GL + Q_GH)

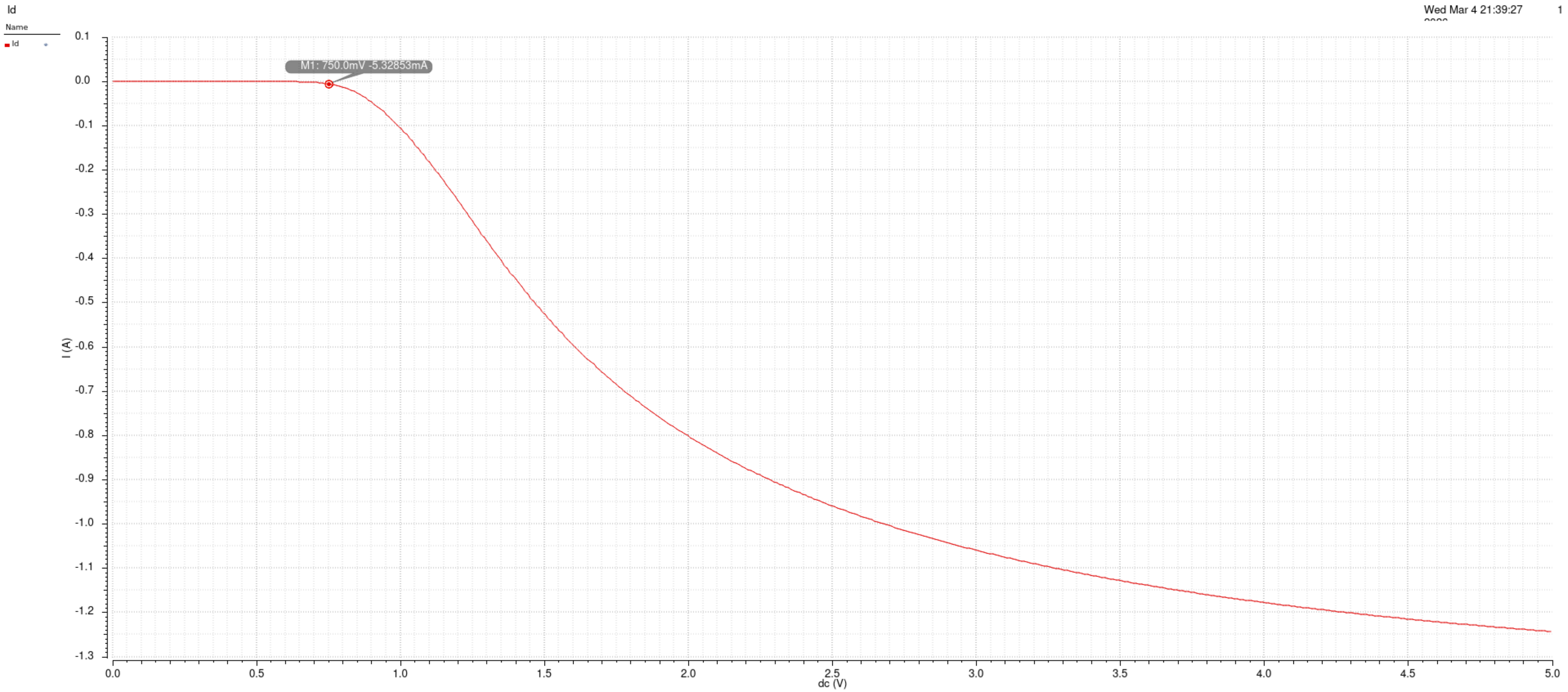
Observations:

Gate current peaks at ~450mA during turn-on

Miller plateau visible in gate voltage ramp

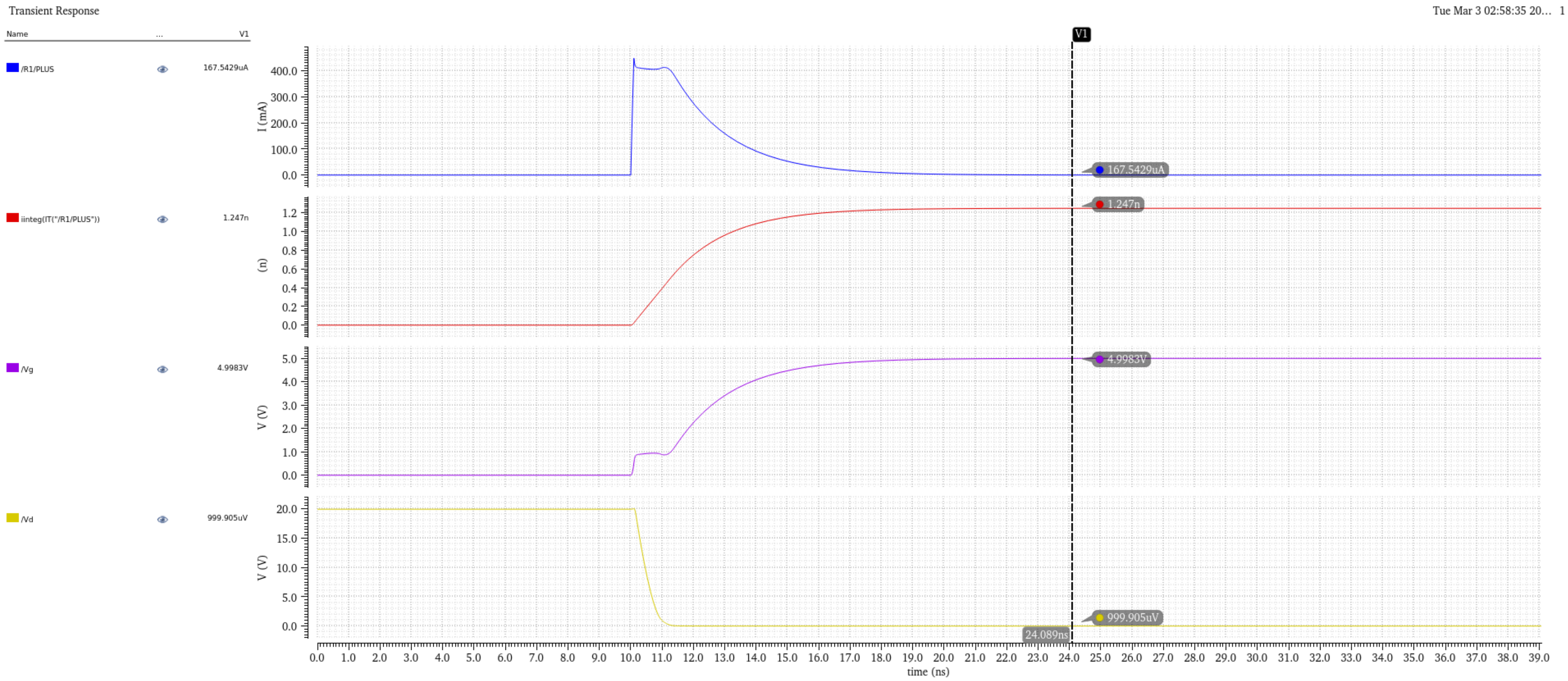
Drain voltage collapses from 20V to ~1mV as FET turns on

Part 2: V_TH Extraction



DC sweep of ID vs VGS. V_TH ~ 0.75V at onset of conduction

Part 2: Gate Charge Q_G



Gate current, integrated charge (Q_G=1.247nC), gate voltage, drain voltage

Part 3

Gate Driver Sizing

Part 3: Gate Driver Sizing

Target: ~25ns rise/fall time at PowerFET gate

Driver sizing:

PMOS: pmos5v_mac, W = 3u, L = 500n

NMOS: nmos5v_mac, W = pmos_w x 4 = 12u, L = 600n

Skewed design (W_NMOS >> W_PMOS):

Fast turn-OFF (strong NMOS pull-down) prevents shoot-through

Slower turn-ON (weaker PMOS pull-up) is acceptable with dead time

Measured results:

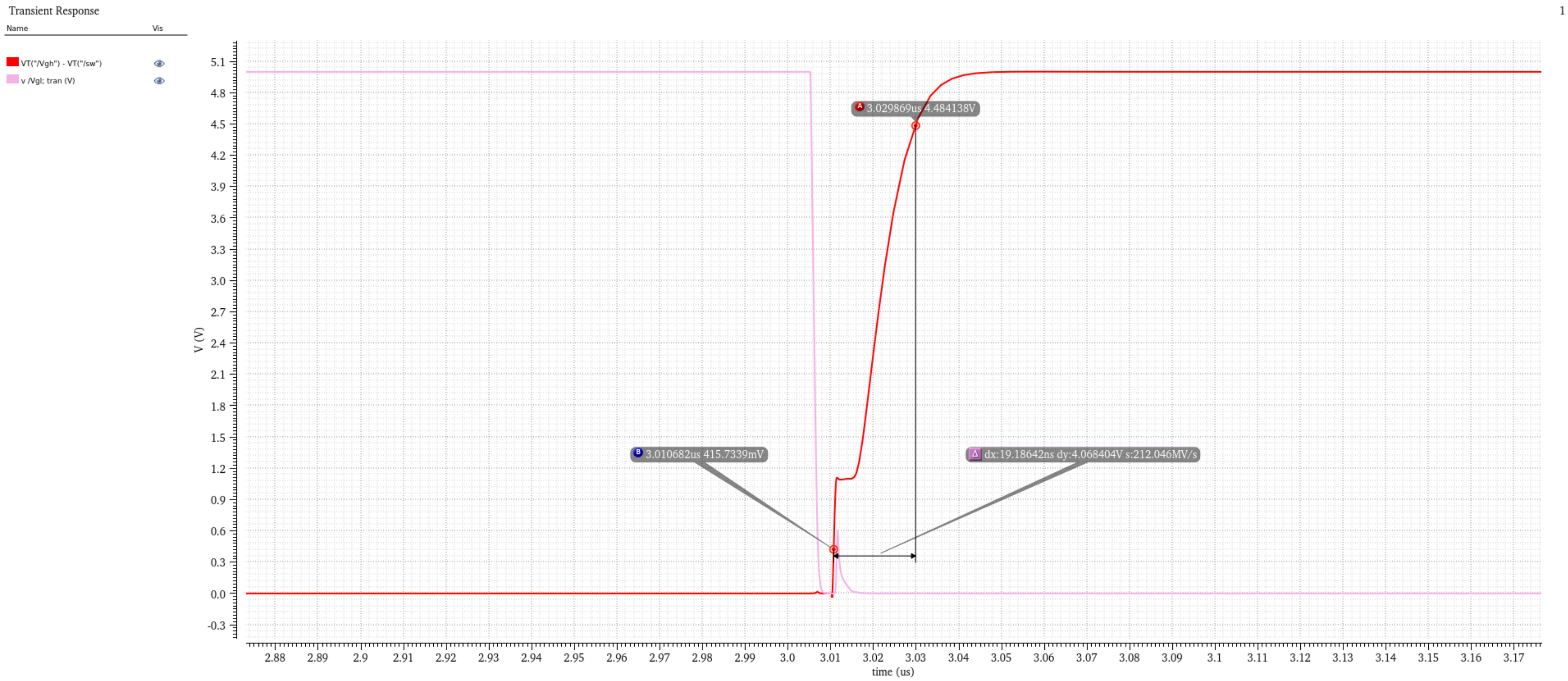
Rise time (Vgl: 0 to 5V): 19.19ns

Fall time (VgsH: 5V to 0V): 2.61ns

Average gate drive current:

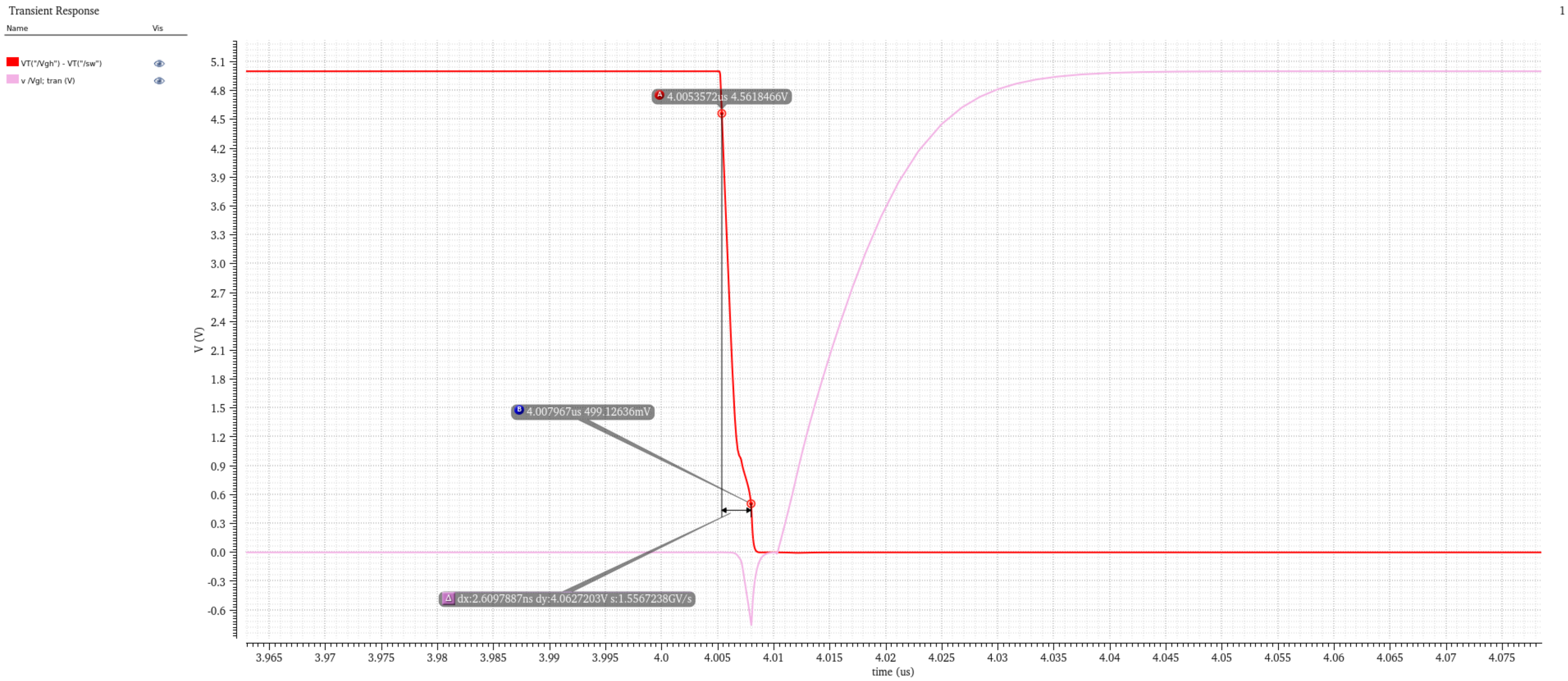
I_avg = Q_G / t_rise = 1.247nC / 25ns ~ 50mA

Part 3: Rise Time



Vgl rise time = 19.19ns (turn-ON of low-side FET)

Part 3: Fall Time



VgsH fall time = 2.61ns (fast turn-OFF via strong NMOS pull-down)

Part 4

Dead Time Verification

Part 4: Dead Time

Dead time = 5ns (break-before-make)

Implementation:

ahdlLib nand_gate + not_gate with tdel = 5ns

Non-overlap generator ensures both FETs are never ON simultaneously

Measurement:

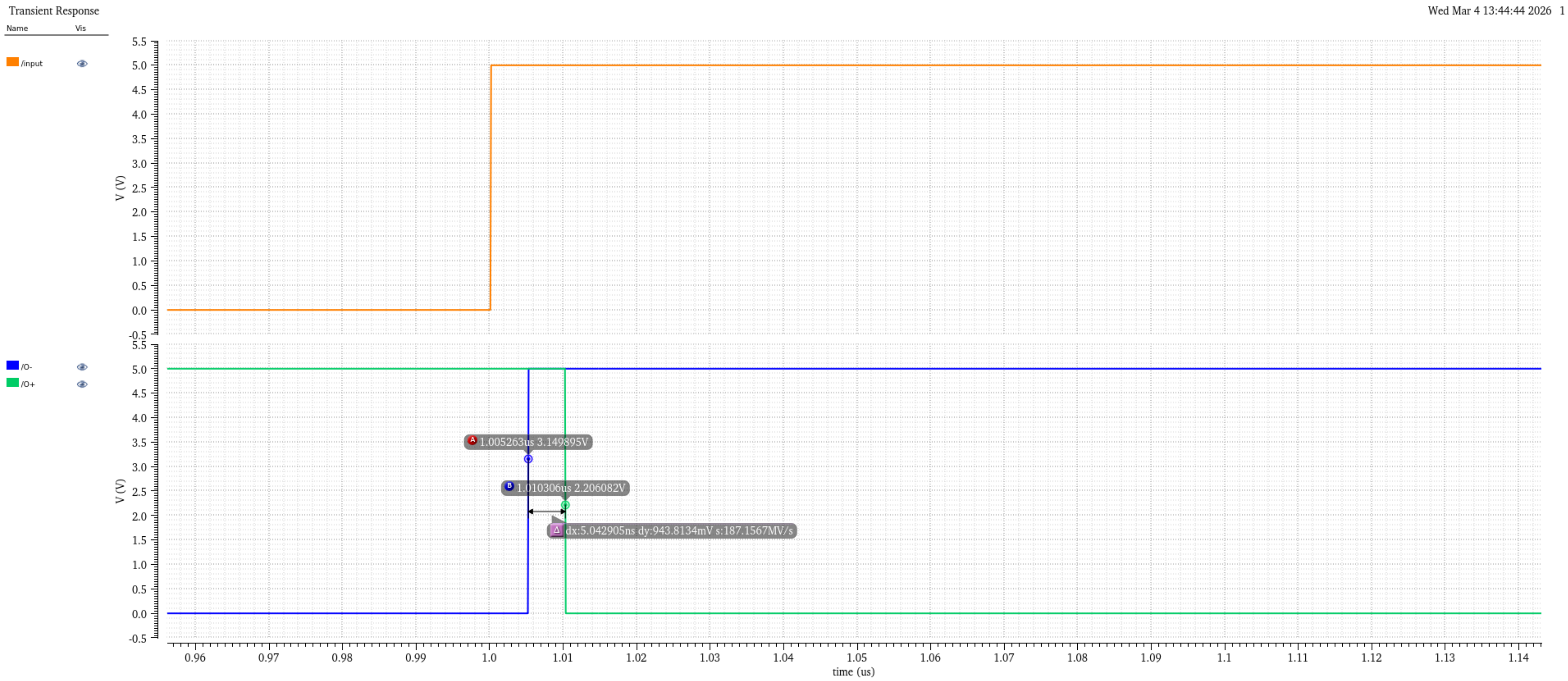
Zoomed to rising edge of /input at ~1.0us

O- falls first, then after 5.04ns, O+ rises

During this gap, both gate drive signals are LOW

Both PowerFETs are OFF (prevents shoot-through)

Part 4: Dead Time Verification



O- falls first, O+ rises 5.04ns later. Both FETs OFF during dead time.

Part 5

Half-Bridge Verification (+/-2A)

Part 5: Soft vs Hard Switching

Soft Switching (iload = +2A):

Load current assists the transition

Faster dVsw/dt, body diode conducts after transition

VgsH stays low during dead time (no shoot-through risk)

Hard Switching (iload = -2A):

Load current resists the transition

Slower dVsw/dt, body diode conducts during dead time

VgsH rises to ~3V due to Cgd coupling (shoot-through risk)

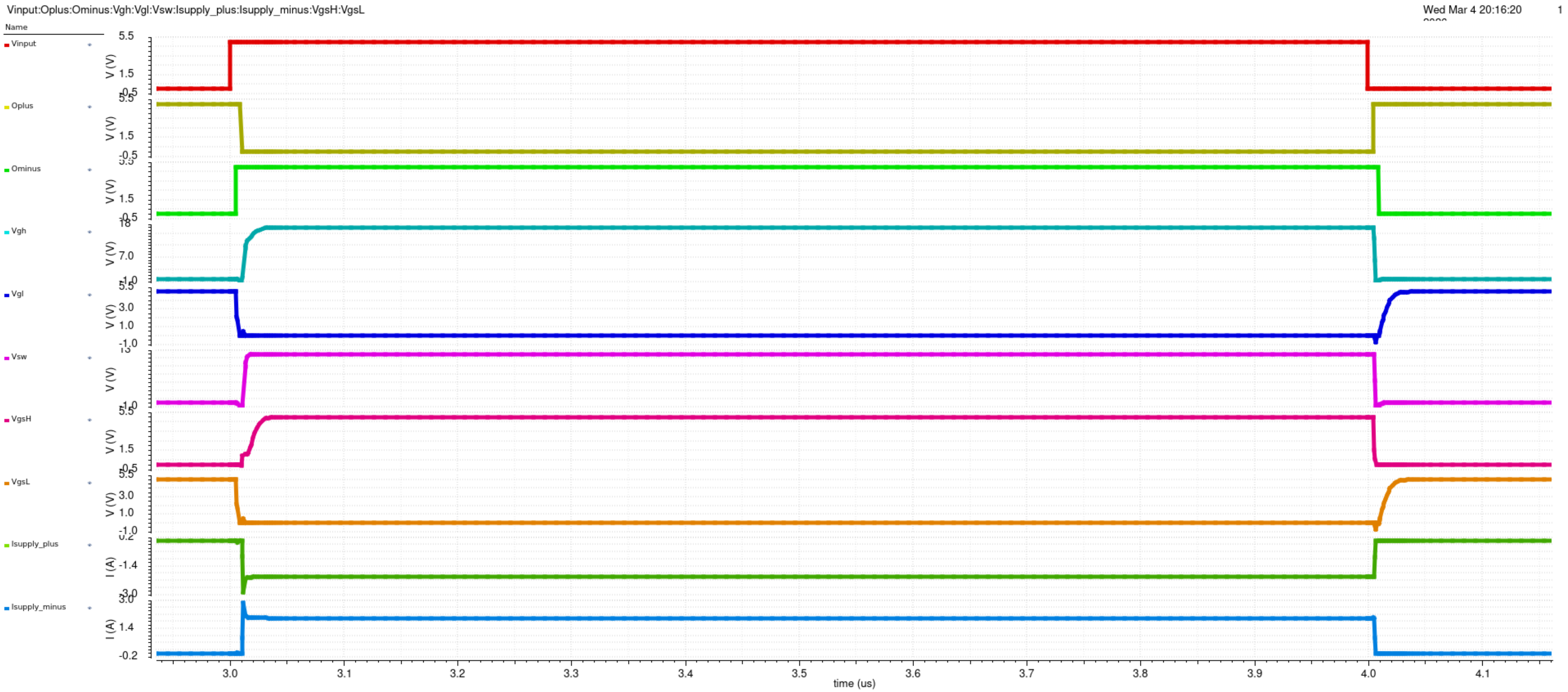
Skewed driver keeps VgsH below V_TH

Cgd coupling formula:

I_GD = C_GD x dV_sw/dt

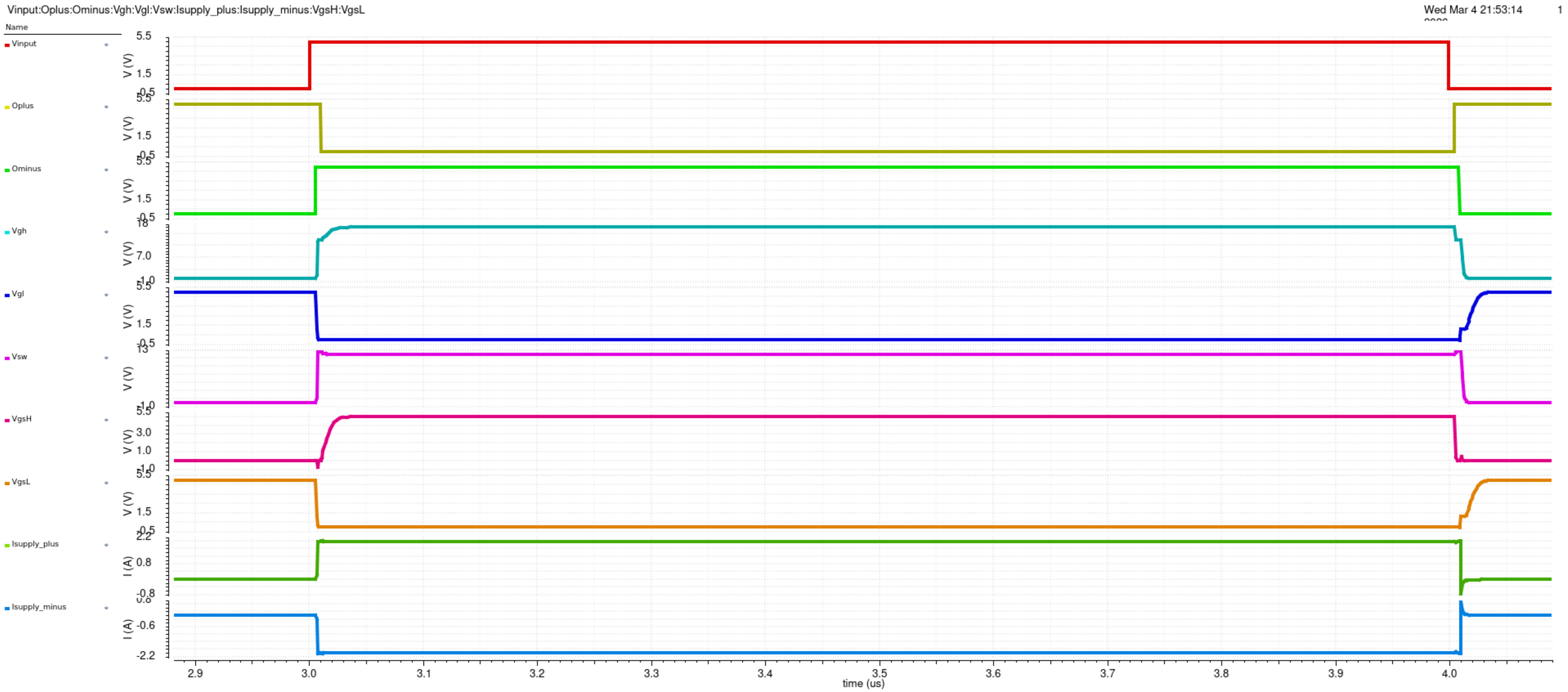
This current pulls V_GS of OFF FET above V_TH -> shoot-through

Part 5: All Signals (+2A, Soft Switching)



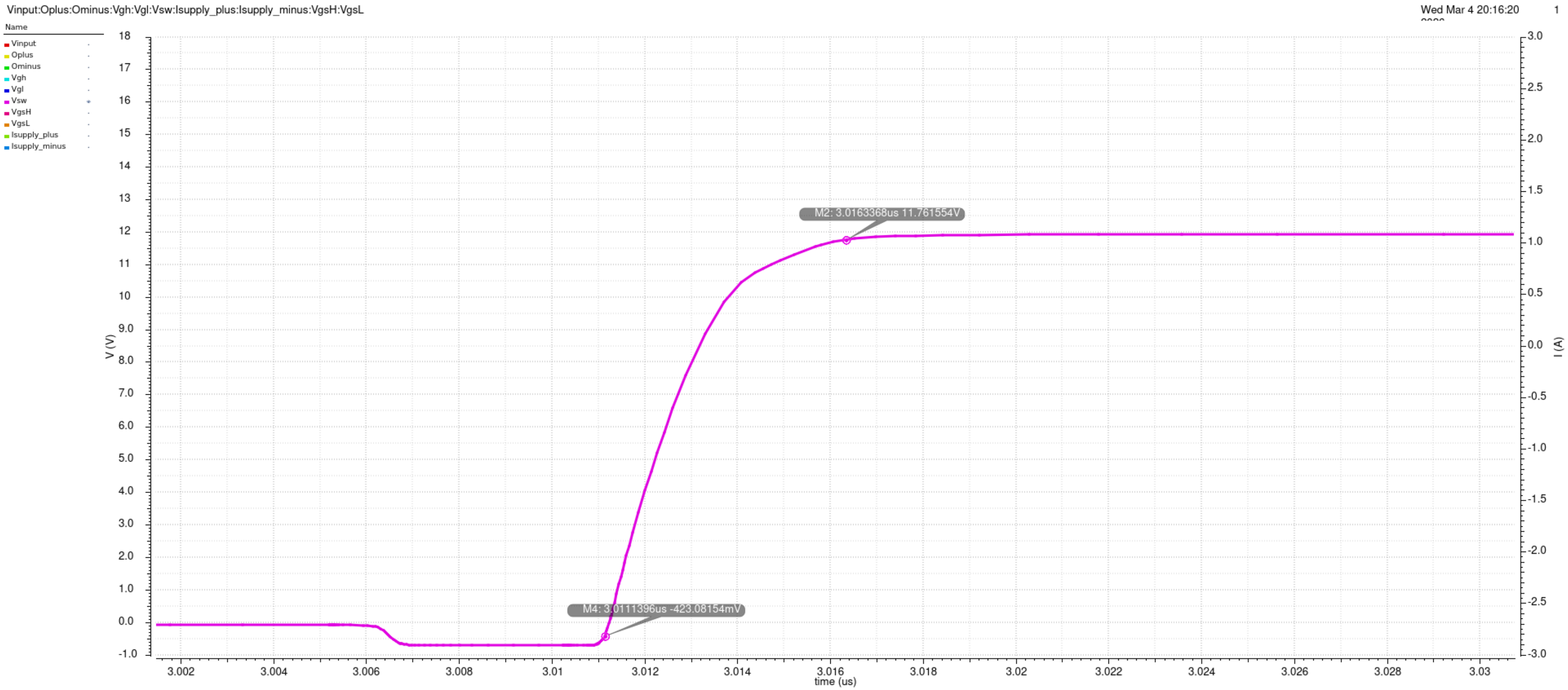
Vinput, O+/O-, Vgh, Vgl, Vsw, VgsH/VgsL, Isupply. Clean switching.

Part 5: All Signals (-2A, Hard Switching)



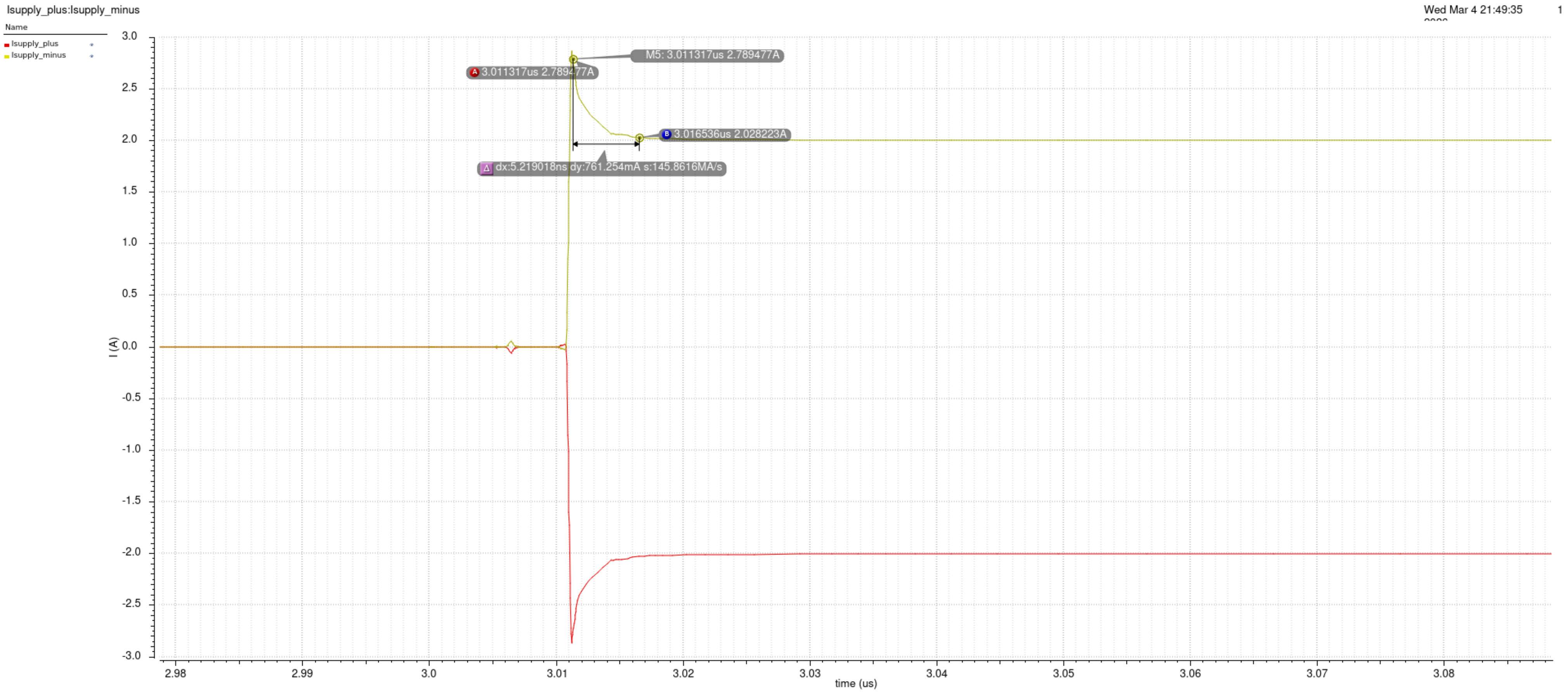
VgsH rises to ~3V during dead time due to Cgd coupling. No shoot-through.

Part 5: Switching Node Zoomed



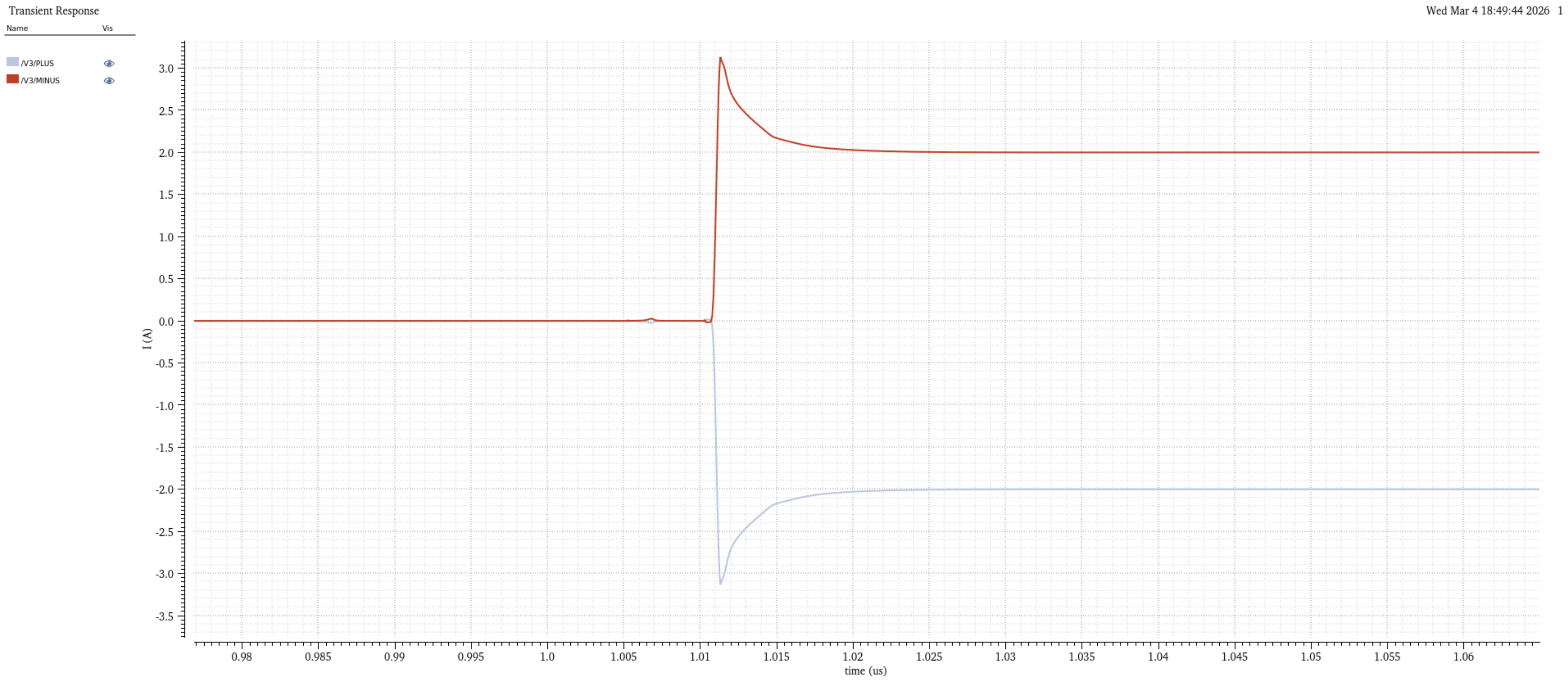
Vsw transition from ~-1V to 12V in ~5ns. Clean, no ringing.

Part 5: Shoot-Through Check



Brief capacitive spike (~2.79A, ~5ns) during transition. No sustained overlap.

Part 5: Load Current at Turn-On



Initial inrush: spikes to ~3.2A, settles to 2A. Normal behavior.

Part 6

BTL Configuration with LC Filter

Part 6: BTL Architecture

Why BTL (Bridge-Tied Load)?

Single half-bridge: sw toggles 0-12V, DC average = 6V offset

BTL: two half-bridges drive load differentially

V_diff = sw_A - sw_B, swings +/-12V with zero DC offset

AD-PWM (Analog Double-Edge):

Half-bridge A: O+ to high-side, O- to low-side

Half-bridge B: O- to high-side, O+ to low-side (swapped)

When A=HIGH, B=LOW and vice versa

LC Filter:

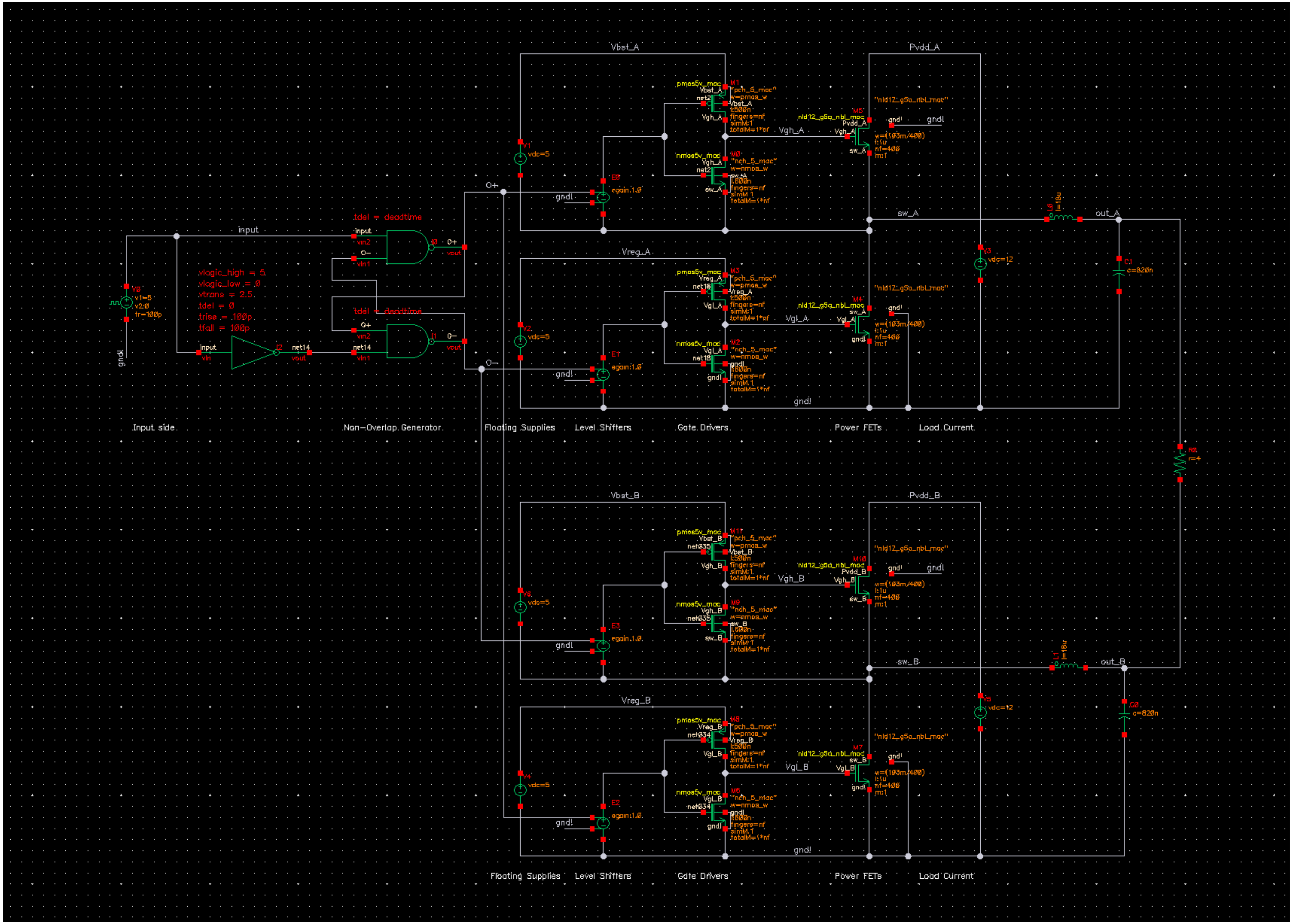
L = 18uH, C = 820nF per side

f0 = 1/(2*pi*sqrt(L*C)) = 41.4 kHz

Passes audio (20Hz-20kHz), blocks 1MHz switching

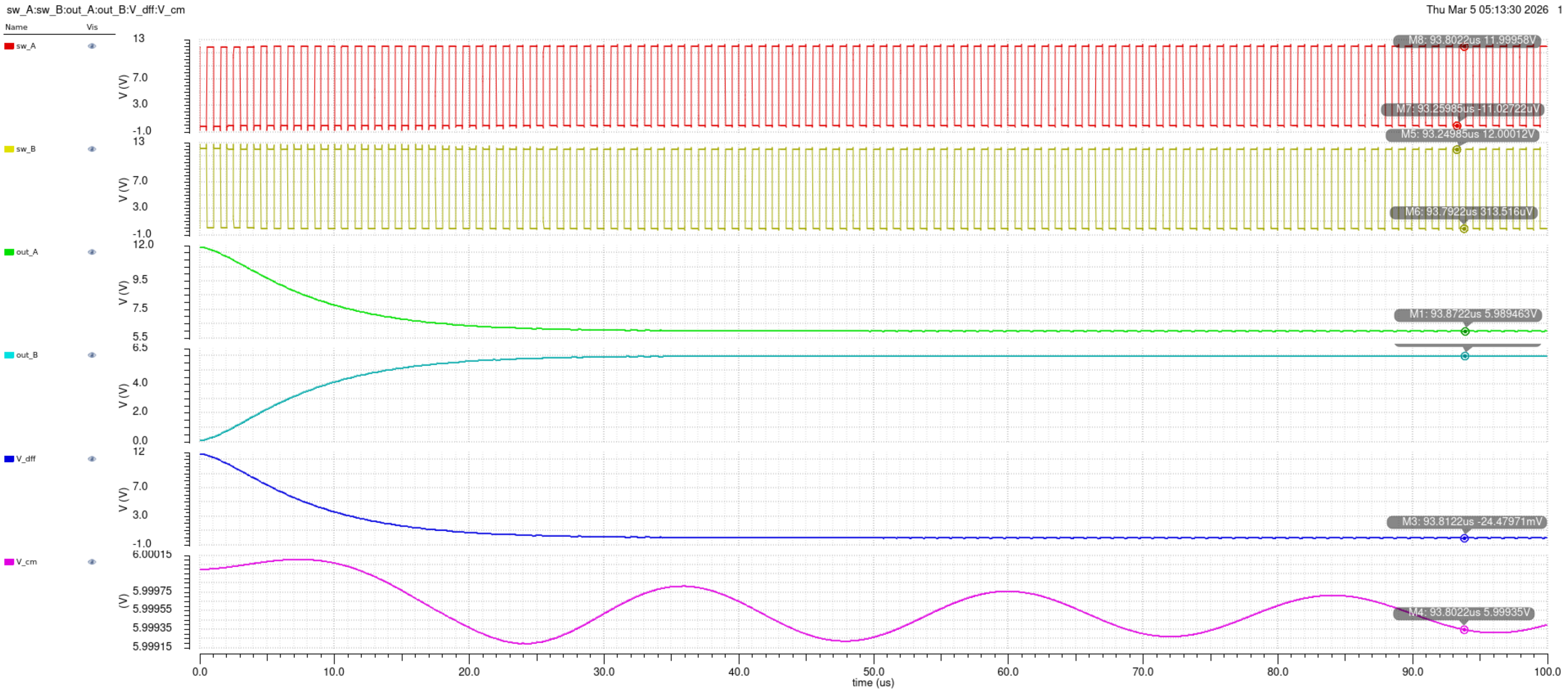
Load: 4 Ohm connected differentially (out_A to out_B)

Part 6: BTL Schematic



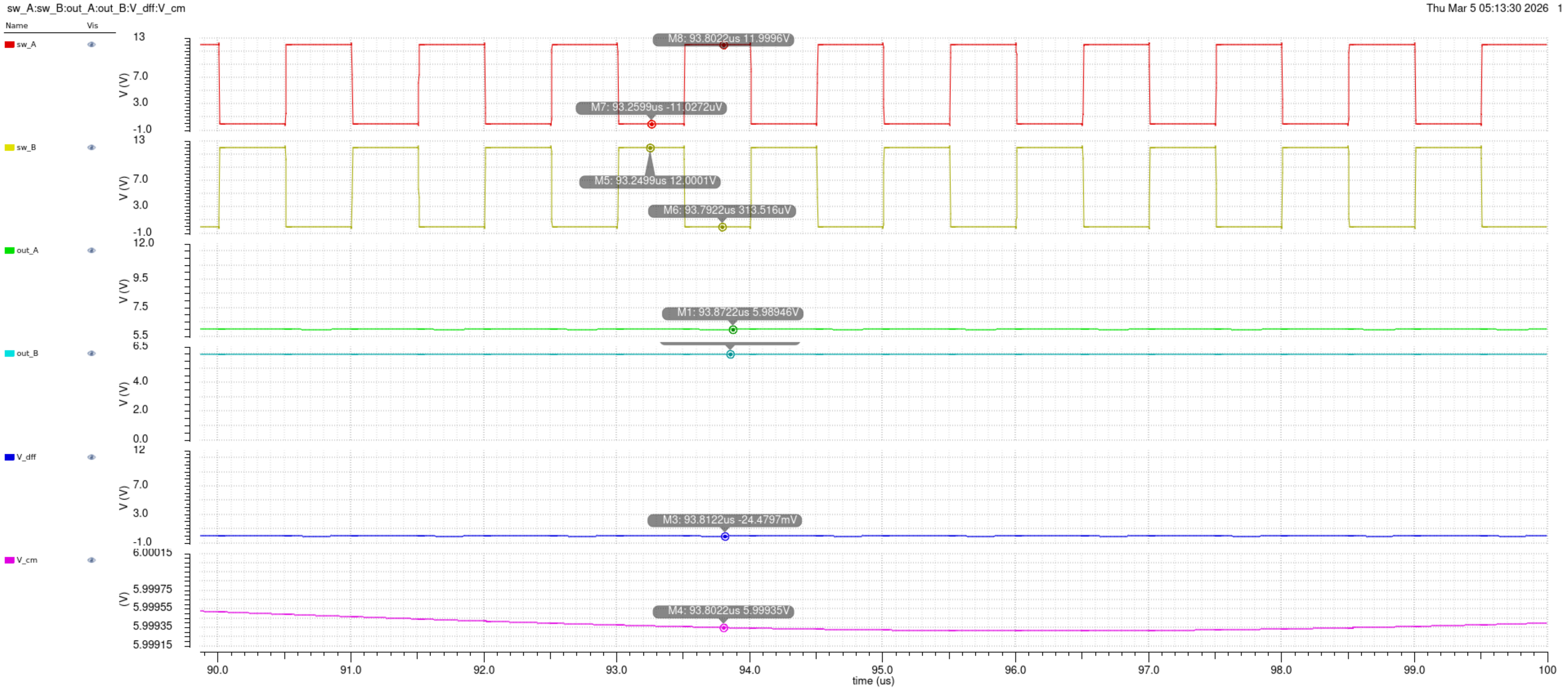
Two half-bridges with swapped O+/O- for AD-PWM, LC filters, differential load

Part 6: BTL at mi=0 (Settling)



Full 100us: sw_A/sw_B inverted, out_A/out_B settle to ~6V, V_diff -> 0V

Part 6: BTL at mi=0 (Settled)



Steady state: V_diff = -24.5mV (~0V), V_cm = 6.0V (constant, AD-PWM property)

Part 6: Efficiency Measurement

Comparator-based PWM for mi sweep:

Triangle carrier (1MHz) + DC signal (mi) + ahdlLib comparator

Duty cycle proportional to modulation index mi

Efficiency formula:

eta = P_load / (P_sup + P_reg)

P_sup = V_SUP x average(I_SUP)   [12V supply]

P_reg = V_REG x average(I_REG)   [5V gate driver supply]

P_load = average(V_diff x I_load)

Parametric sweep: mi = 0 to 0.9 (10 steps)

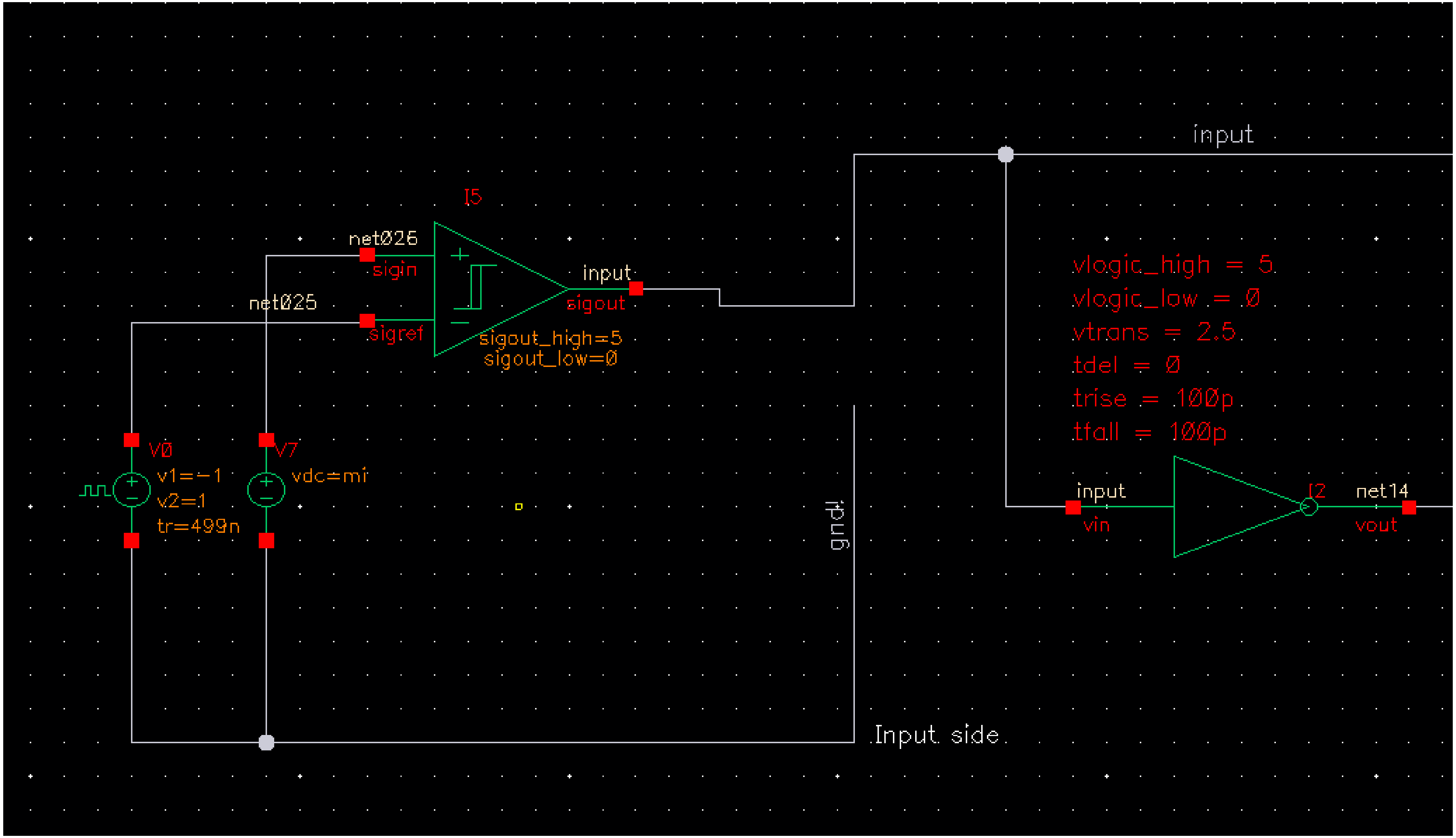
Power loss breakdown:

P_R (conduction) = I^2 x R_ON -- increases with mi

P_Q (gate charge) = f_SW x 2 x V_SUP x Q_G -- constant

P_X (switching) = f_SW x t_x x V_SUP x I_OUT -- hard switching only

Part 6: PWM Input Stage



Triangle carrier + DC signal (mi) + comparator generates variable-duty PWM

Part 6: Efficiency vs mi (1x Drivers)

P_su, P_reg, P_load, eta vs mi. Efficiency increases with output power.

Part 7

4x Gate Driver Comparison

Part 7: 1x vs 4x Gate Drivers

Tradeoff:

Larger driver -> faster switching -> less P_X (switching loss)

Larger driver -> more driver capacitance -> more P_Q (gate charge loss)

1x Drivers: pmos_w = 3u, nmos_w = 12u

4x Drivers: pmos_w = 12u, nmos_w = 48u

Expected behavior:

Low mi: P_Q dominates -> 4x drivers = worse efficiency

High mi: P_X dominates -> 4x drivers = better efficiency

Crossover point where bigger drivers become beneficial

Simulation result:

1x and 4x efficiency curves are very similar

P_reg difference only ~0.3mW (20.2 vs 20.5mW)

Due to ideal components (vcvs level shifters, ahdlLib gates)

Real implementation would show larger differences

Part 7: Efficiency vs mi (4x Drivers)

4x drivers (pmos_w=12u): nearly identical efficiency to 1x drivers

Part 7: 1x vs 4x Comparison

Parameter	1x (pmos_w=3u)	4x (pmos_w=12u)
P_reg (gate driver)	~20.2 mW	~20.5 mW
P_load at mi=0.9	~28 W	~28 W
Switching speed	Baseline	~4x faster
Efficiency curve	Baseline	Nearly identical

Summary

Part 1: PowerFET Sizing

nf=400 fingers, W=103mm, R_ON=50.02 mOhm at 150C

Part 2: V_TH = 0.75V, Q_G = 1.247 nC

Part 3: Gate Driver - Rise=19ns, Fall=2.6ns (skewed: W_N=4xW_P)

Part 4: Dead Time = 5.04ns (break-before-make verified)

Part 5: Half-Bridge verified at +/-2A

Soft switching: clean. Hard switching: VgsH rises to 3V but no shoot-through

Part 6: BTL with LC Filter

V_diff~0V at mi=0, V_cm=6V constant (AD-PWM property)

Efficiency increases with modulation index (output power)

Part 7: 4x drivers show similar efficiency (ideal component limitation)